Jason Hernandez Lopez

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EXPERIENCE

Build Engineer Intern

Albedo **May 2026 - August 2026, Broomfield, CO**

- Authored the company's official 134-page Rev1 satellite bus build guide, defining the manufacturing sequence with build photos, CAD views, part/tool callouts, warnings, and verification steps.
- Designed and validated mechanical tooling for satellite bus assembly, including 4 CAD indexers released as production parts and steel/PETG caster adapters used to retrofit four 1,100-lb motorcycle jacks.
- Troubleshot a thermal-vacuum test system, restoring vacuum-chamber venting and isolating a separate heater-power fault for vendor repair.

Hardware Intern

NewHaptics **January 2026 - April 2026, Ann Arbor, MI**

- Designed and built hardware for an automated imaging gantry system through motor selection, lead-screw mechanism, baseplate design, machining, assembly, and test-script development.
- Built and tested electromechanical braille-module prototypes; designed hardware test setups and scripts to evaluate reliability, lifecycle performance, and prototype behavior.

Robotics Hardware Intern

Serve Robotics **May 2025 - August 2025, Miami, FL**

- Scaled an operational robot fleet from 15 to 30 in 4 weeks and to 75 by week 10, averaging 10–12 electromechanical repairs/week across actuators, sensors, and PCBs using oscilloscopes, multimeters, and machine shop tools.
- Sustained ≥65% fleet availability in the first 4 weeks and ~80% over the next 8 weeks by designing, installing, and testing drivetrain reinforcements and mechanical fixtures.

Robot Technician

Starship Technologies **January 2025 - April 2025, Detroit, MI**

- Served as the sole technician for 4 L4 autonomous robots, independently owning daily operation, maintenance, and electromechanical troubleshooting across drivetrain, power, and sensors.
- Performed end-to-end mechanical/electrical repairs, robot upgrades, parts/tooling management, and service documentation in JIRA/Confluence.

PROJECTS

- **STRYKER** – 6-DOF Robotic Arm · Mechanical Design · Electrical Design · Software Design
- **SENTINEL** – Self-Balancing Ballbot · Mechatronics · Embedded Systems · Control System Design

EDUCATION

Bachelor of Science in Robotics - Hardware and Sensors Concentration

University of Michigan College of Engineering · Ann Arbor, MI · May 2027 · GPA: 3.6

SKILLS

Mechanical: SolidWorks, Creo, Fusion 360, GD&T, DFM/DFA, Mechanical Drawings, Machining, Rapid Prototyping.

Electrical/Test: Altium, EasyEDA, LTspice, PCB Assembly/Rework, Soldering, Wiring, Oscilloscope, Multimeter

Programming/Controls: MATLAB/Simulink, Python, C++, Embedded Systems, AI-Assisted Development